Roll Number: _____

Amrita Vishwa Vidyapeetham

Mid - Term Examination, October 2025

Third Semester

Int. M. Sc (Mathematics & Physics)

24MAT208 Introduction to Linear Algebra

Scientific calculators are allowed

Duration :2 hours

Maximum : 50 Marks

Answer all questions.

Qn No.	Questions	Marks	CO	BTL
1	If the rank of the matrix $A = \begin{bmatrix} 5 & -3 \\ k & 6 \end{bmatrix}$ is 1, then the value of k is?	2	CO3	L2
2	Consider the vectors $u = (1, -3, 2)$ and $v = (2, -1, 1)$ in $\mathbb{R}^3$. Find k so that $(1, k, 5)$ is a linear combination of u and v.	2	CO2	L1
3	Find a basis and dimension of the subspace, $W = \{(x, y, z) \mid 2x + y + z = 0\}$ of $\mathbb{R}^3$.	2	CO3	L2
4	Let $f(x) = \cos^2 x$ and $g(x) = \sin^2 x$. Which of the following functions lie in the space spanned by f and g? Justify your answer. (a) $\cos 2x$ (b) 2 (c) $\sin x$ (d) 0.	4	CO4	L3
5	Investigate the values of α and β, so that the equations, $2x + 3y + 5z = 9$, $7x + 3y - 2z = 8$ and $2x + 3y + \alpha z = \beta$ have: (a) no solution (b) a unique solution (c) an infinite number of solutions.	6	CO1	L2
6	Find one vector in $\mathbb{R}^3$ which spans the intersection of U and W, where U is the xy – plane and W is the space spanned by the vectors $(1, 1, 1)$ and $(1, 2, 3)$.	6	CO3	L3
7	Check whether W is a subspace of the given vector space V with respect to the standard operations.	6	CO3	L1

	(a) V is vector space of all 2×2 matrices with real entries, W is the set of all 2×2 matrices of the form $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, where $ad = 0$. (b) $V = \mathbb{R}^3$, $W = \{(x, y, z) \in \mathbb{R}^3 \mid z = 0,\ x - y + z - 1 = 0\}$.			
8	Find a basis for the solution space of the following homogeneous linear system and hence find the dimension of that space. $x + 2y - z + w = 0$ $2x + 4y - 2z + 2w = 0$. $-3x - 6y + 3z - 5z = 0$	**6**	**CO4**	**L3**
9	Find the rank for the matrix by Gauss elimination: $A = \begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 3 & 3 & 2 \\ 3 & 2 & 1 & 3 \\ 6 & 8 & 7 & 5 \end{bmatrix}$.	**8**	**CO2**	**L2**
10	Are the following vectors linearly dependent? If so, find a relation between them: $V_1 = \begin{bmatrix} 1 \\ 4 \\ 2 \\ 7 \end{bmatrix}, V_2 = \begin{bmatrix} 2 \\ 1 \\ 3 \\ 6 \end{bmatrix}, V_3 = \begin{bmatrix} 1 \\ 2 \\ -3 \\ 0 \end{bmatrix}, V_4 = \begin{bmatrix} 3 \\ -1 \\ 1 \\ 3 \end{bmatrix}$	**8**	**CO2**	**L2**

L_1-Remember, L_2-Understand, L_3-Apply, L_4-Analyze, L_5-Evaluate, L_6-Create